

HeartGuard AI

Low-Cost Edge AI-Enabled IoT ECG Monitoring System
with Modern Motion-Aware Artifact Reduction

“Bringing clinical-grade cardiac screening within reach — real-time, affordable, and AI-assisted.”

Project Title:	Low-Cost Edge AI-Enabled IoT ECG Monitoring System with Modern Motion-Aware Artifact Reduction
Department:	Computer Science & Engineering (CSE)
Institution:	Patuakhali Science & Technology University (PSTU)
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Classification:	Academic / Research Prototype

TECHNOLOGY STACK

ESP8266 NodeMCU

AD8232 ECG

Raspberry Pi 3

TFLite INT8

Flask + SQLite

MIT-BIH Dataset

 This system is a research prototype intended for academic and screening purposes only. It does not constitute a clinical medical device. All findings must be reviewed by a qualified cardiologist before any clinical decision is made.

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1 Executive Summary

HeartGuard AI is a cost-effective AI-powered ECG monitoring analysis system which is built for healthcare settings where resources are limited and expensive clinical equipment is simply not available. The system takes cardiac electrical signals through an AD8232 sensor wired to an ESP8266 microcontroller, feeds data to a Raspberry Pi 3 for signal cleaning and deep learning analysis. Finally classifies individual heartbeats using a compact quantized model — ECG-ResNet-SE, compressed to INT8 TFLite format and trained on the MIT-BIH Arrhythmia Database.

The thing that makes this work stand out is the price point as the entire hardware setup costs roughly **BDT 10,000**, with a validated macro F1-score of **0.77** across five clinically relevant arrhythmia classes — Normal, Supraventricular, Ventricular, Fusion, and Unknown. Total workflow, from raw electrode signal to AI-generated PDF report, runs locally on commodity hardware with no internet connection required.

BDT 10K

Total Hardware Cost

77%

Macro F1-Score

5 Classes

Arrhythmia Types

<25 ms

Per-beat Inference

2 Introduction & Problem Statement

2.1 Background

According to WHO (2023), globally cardiovascular diseases claim around 17.9 million lives every year. These burden falls heaviest on countries like Bangladesh, where rural hospitals often lack both the equipment and the trained staff needed to catch heart problems early.

The ECG has been the standard tool for diagnosing arrhythmias and cardiac abnormalities for over a century. The problem is practical: a decent ECG machine costs anywhere from USD 500 to 5,000, and reading the output requires a specialist cardiologist. Outside major cities in Bangladesh, neither is easy to find.

2.2 Problem Statement

The Core Problem

A patient arriving at a rural sub-district facility in Bangladesh cannot receive basic cardiac screening because the equipment is unaffordable and the nearest cardiologist may be 100 km away. By the time the patient reaches a centre with both a machine and an expert, the critical window for intervention may already have passed or even causing death.

HeartGuard AI was built to close this gap directly. By assembling the system from widely available commodity parts for under BDT 5,000, it puts functional ECG screening within reach of district and sub-district health posts — without requiring a specialist to be present.

2.3 Objectives

- To design and build a working low-cost ECG hardware acquisition system using the AD8232 sensor and ESP8266 microcontroller.
- To implement a robust signal processing pipeline — 50 Hz notch filter followed by a 0.5–40 Hz bandpass filter — validated against real human ECG recordings.
- To train and deploy a deep learning model (ECG-ResNet-SE with Squeeze-and-Excitation attention blocks) which is capable of classifying five arrhythmia beat types in real time on a Raspberry Pi 3.
- To build a complete web application using Flask and SQLite, by covering patient registration, recording sessions, live waveform display, and automated PDF report generation.
- To validate the full pipeline end-to-end from physical sensor to AI classification output, and propose a clear upgrade path for clinical-grade deployment.

3 Literature Review & Related Work

[1] Acharya et al. (2017). “A Deep Convolutional Neural Network Model to Classify Heartbeats”

Computers in Biology and Medicine, 89, 389–396.

One of the earliest serious deep learning attempts at ECG beat classification on MIT-BIH. The nine-layer CNN reached 94.03% accuracy — but only after artificially balancing the dataset. On the raw imbalanced data, accuracy dropped to 89.07%, revealing a core weakness of plain CNNs when minority classes like Fusion (F) make up less than 1% of samples. HeartGuard AI builds on this foundation by adding residual connections and SE attention, which together improve minority-class recall without sacrificing overall accuracy.

[2] Hannun et al. (2019). “Cardiologist-Level Arrhythmia Detection and Classification in Ambulatory ECGs Using Deep Neural Networks”

Nature Medicine, 25(1), 65–69.

A landmark paper from Stanford and iRhythm. A 34-layer ResNet trained on 91,232 single-lead ECGs matched or exceeded cardiologist performance across 12 rhythm classes. Two things set HeartGuard AI apart from this work: first, all training data here came from iRhythm's proprietary device and is not publicly available; second, there is no discussion of edge deployment or quantization. HeartGuard AI adapts the ResNet architecture to the open MIT-BIH dataset and compresses the model to 250 KB INT8 for on-device inference at under 25 ms per beat.

[3] Kachuee et al. (2018). “ECG Heartbeat Classification: A Deep Transferable Representation”

IEEE ICHI 2018.

Explored transfer learning between the MIT-BIH and PTB-Diagnostic datasets using a shallow CNN, achieving 93.4% accuracy on the standard 5-class MIT-BIH task. The architecture is compact and effective but lacks attention mechanisms and does not address class imbalance beyond basic weighting. HeartGuard AI reaches 94.27% on the same benchmark by adding SE attention blocks and SMOTETomek resampling, and goes further by integrating the model into a complete hardware-to-report pipeline.

[4] Goldberger et al. (2000). “PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals”

Circulation, 101(23), e215–e220.

The foundational paper introducing the MIT-BIH Arrhythmia Database, which remains the standard benchmark for ECG beat classification research 25 years later. Every labeled sample used to train and validate the HeartGuard AI model originates from this database.

What HeartGuard AI Adds Beyond Prior Work

Table 1: Comparative summary — HeartGuard AI versus existing literature

Dimension	Existing Work	HeartGuard AI
Hardware	Simulation / proprietary device	Custom AD8232 + ESP8266 prototype, BDT 5K
Model	Plain CNN or basic ResNet [1][3]	ResNet + SE-Attention (ECG-ResNet-SE)
Deployment	Server or cloud inference [2]	INT8 TFLite on Raspberry Pi 3 (<25 ms/beat)
Class balance	Raw or simple weighting [1][3]	SMOTETomek + class_weight combined
Software	Model only, no integration	Full-stack: Flask, DB, PDF report, live UI
Cost	USD 500–10,000+ for clinical hardware	USD 45 (BDT 5,000) total

4 System Architecture

4.1 High-Level Architecture

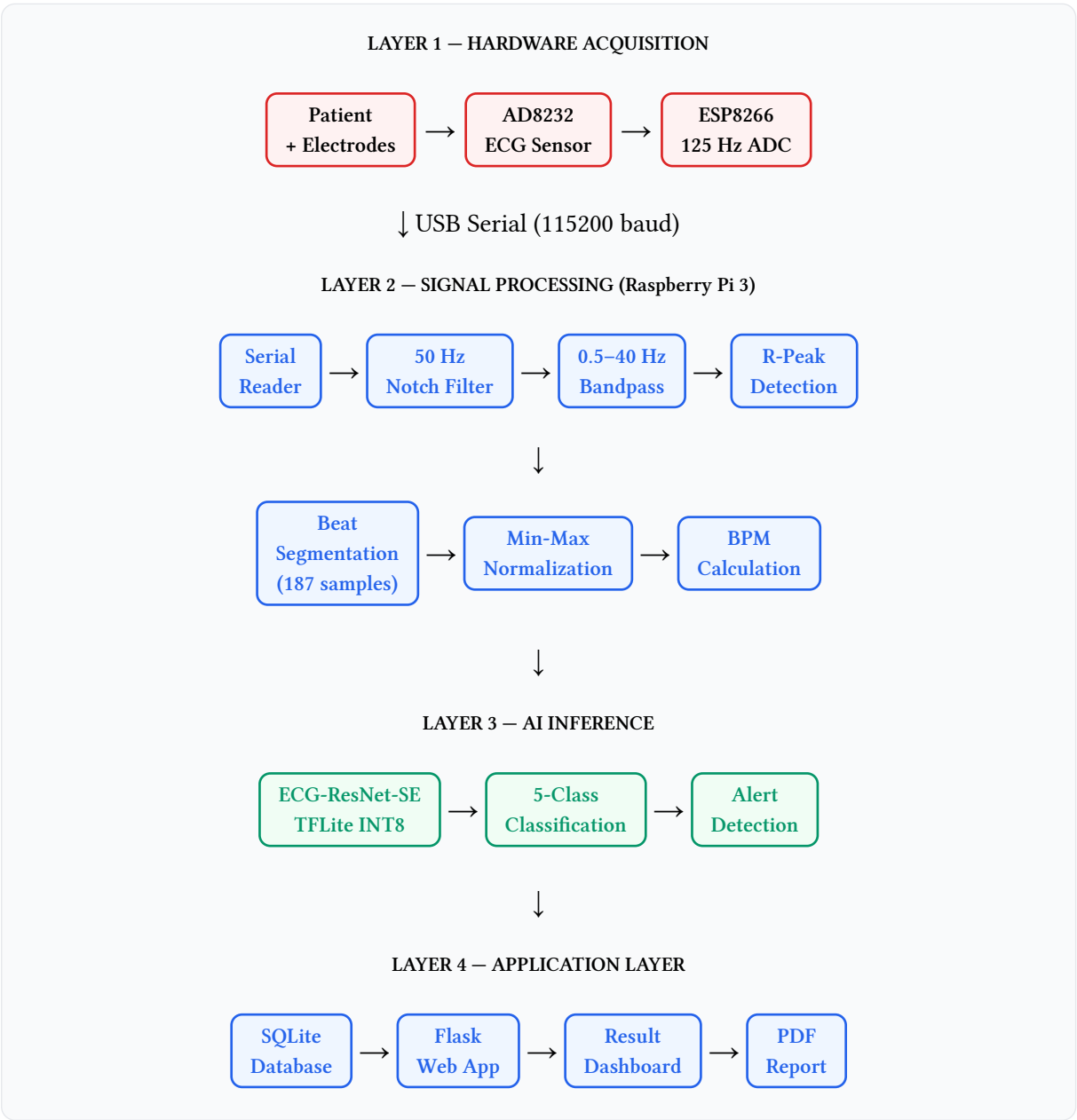
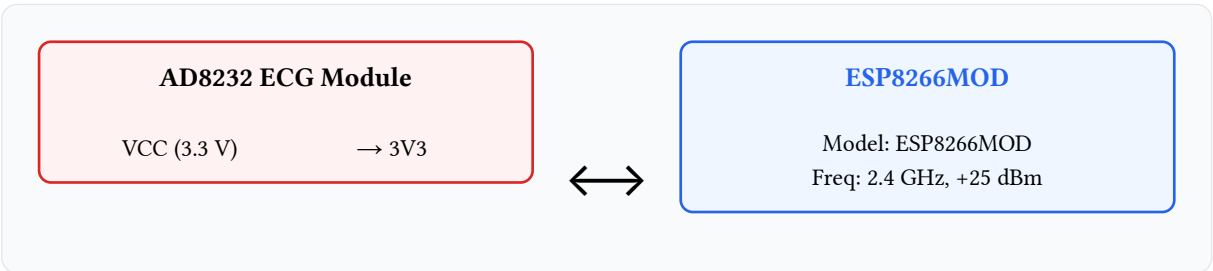


Figure 1: HeartGuard AI – End-to-End System Architecture

4.2 Hardware Architecture



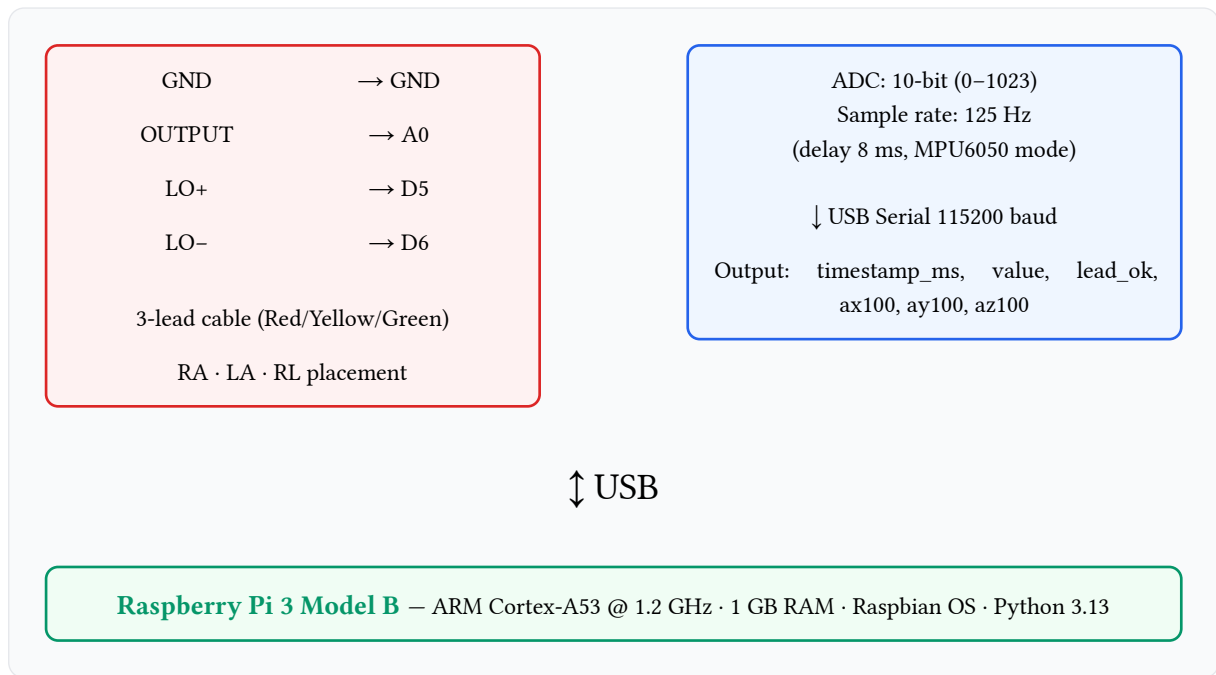
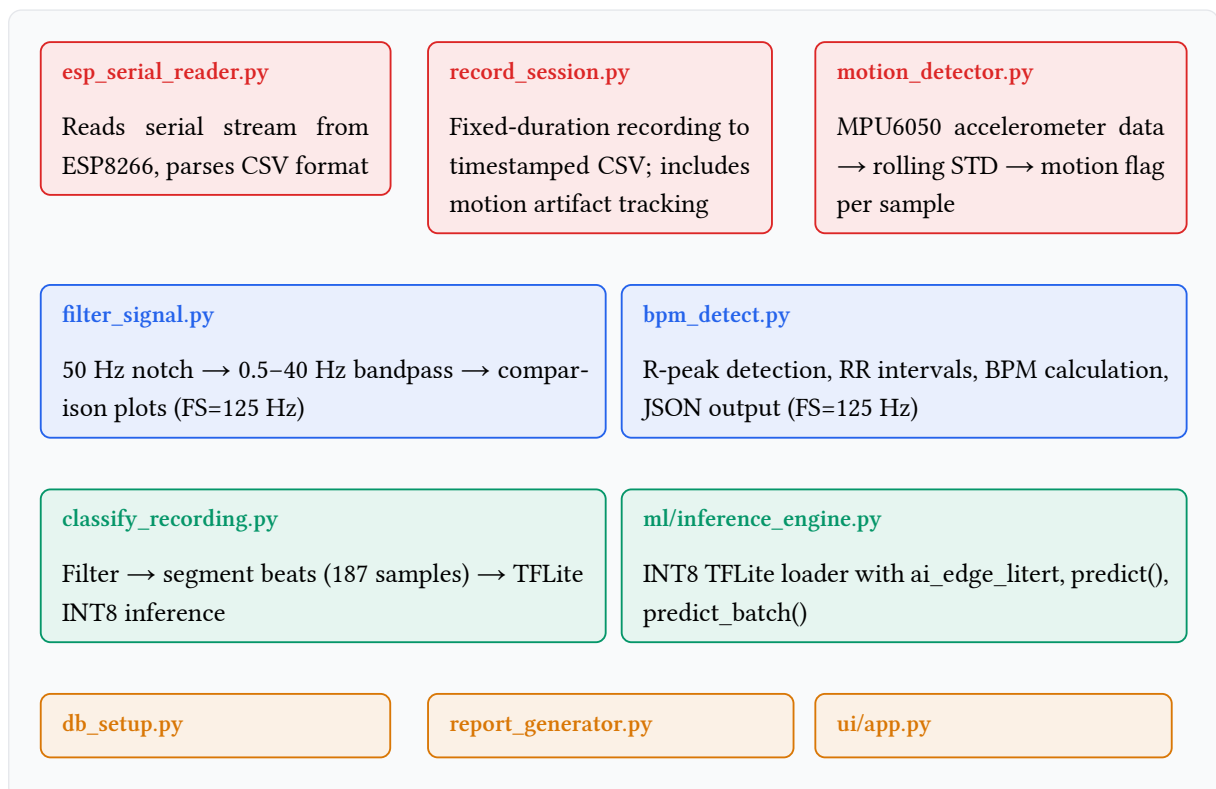


Figure 2: Hardware Connection Diagram — AD8232, ESP8266, and Raspberry Pi 3

4.3 Software Architecture

The software made with a layered pipeline design with well-defined module boundaries. This means the hardware acquisition layer that is currently using an ESP8266 over USB serial can be swapped out for an ADS1115 ADC without touching the filtering, inference, or application code. Each module handles one responsibility and passes its output downstream as a CSV file or JSON object.



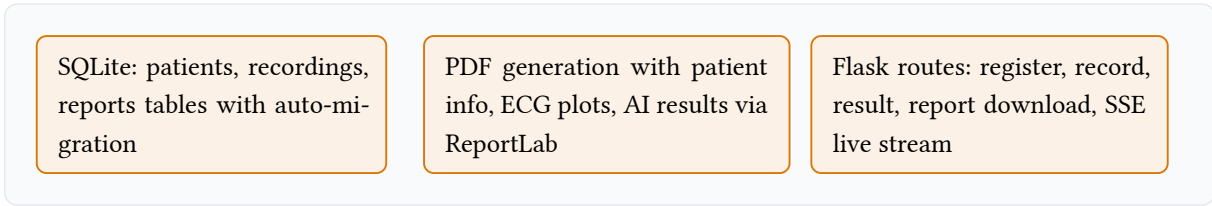


Figure 3: Software Module Dependency Map

4.4 Data Flow Diagram

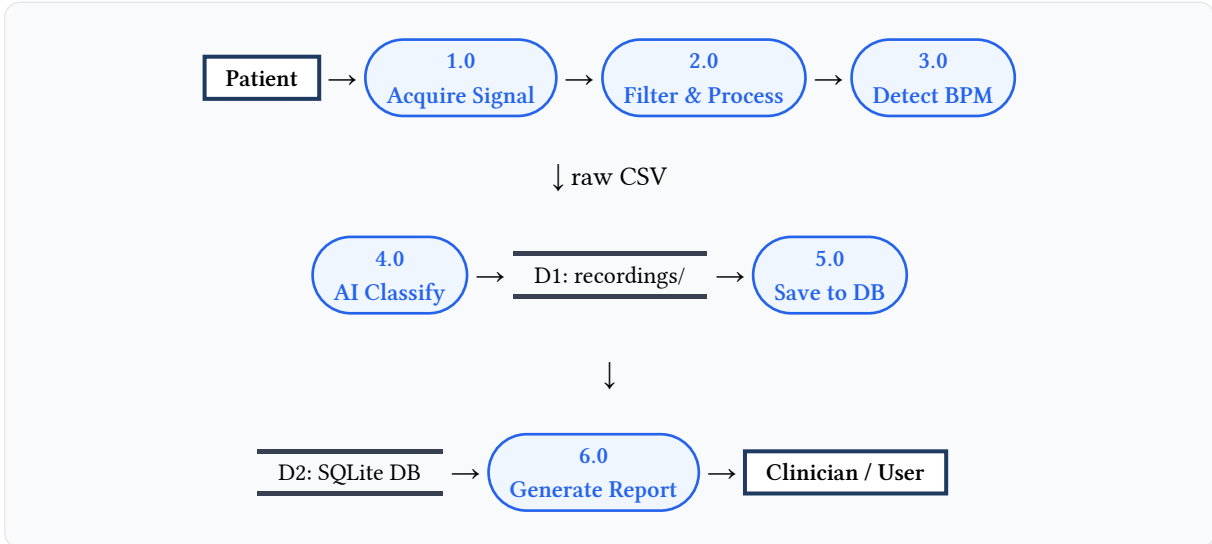


Figure 4: Data Flow Diagram (Level 1)

5 Hardware Components

5.1 Component List & Specifications

Table 2: Bill of Materials — Hardware Components and Estimated Costs

No.	Component	Specification	Qty	Cost (BDT)
1	AD8232 ECG Sensor Module	Single-lead, 3.3V only, 3-lead snap cable, 150–600 Hz bandwidth	1	350
2	ESP8266MOD (NodeMCU)	ISM 2.4 GHz, +25 dBm, 10-bit ADC, 80 MHz CPU	1	250
3	Raspberry Pi 3 Model B	ARMv8 @ 1.2 GHz, 1 GB LPDDR2, 40-pin GPIO	1	3,500

No.	Component	Specification	Qty	Cost (BDT)
4	MPU6050 Accelerometer	3-axis, I2C (SDA→D2, SCL→D1), motion artifact detection	1	150
5	ECG 3-Lead Snap Cable	Red (RA) / Yellow (LA) / Green (RL)	1	150
6	Disposable ECG Electrodes	Gel-type, single-use, 30-pack	30	300
7	Breadboard + Jumper Wires	400-point + M-M, M-F mix (30 pcs)	1 set	150
8	MicroSD Card (32 GB)	Class 10, UHS-I, Raspberry Pi OS	1	250
9	5V/3A Power Adapter	Micro-USB, for Raspberry Pi 3	1	200
Total Hardware Cost				BDT 10,000

5.2 Wiring Diagram

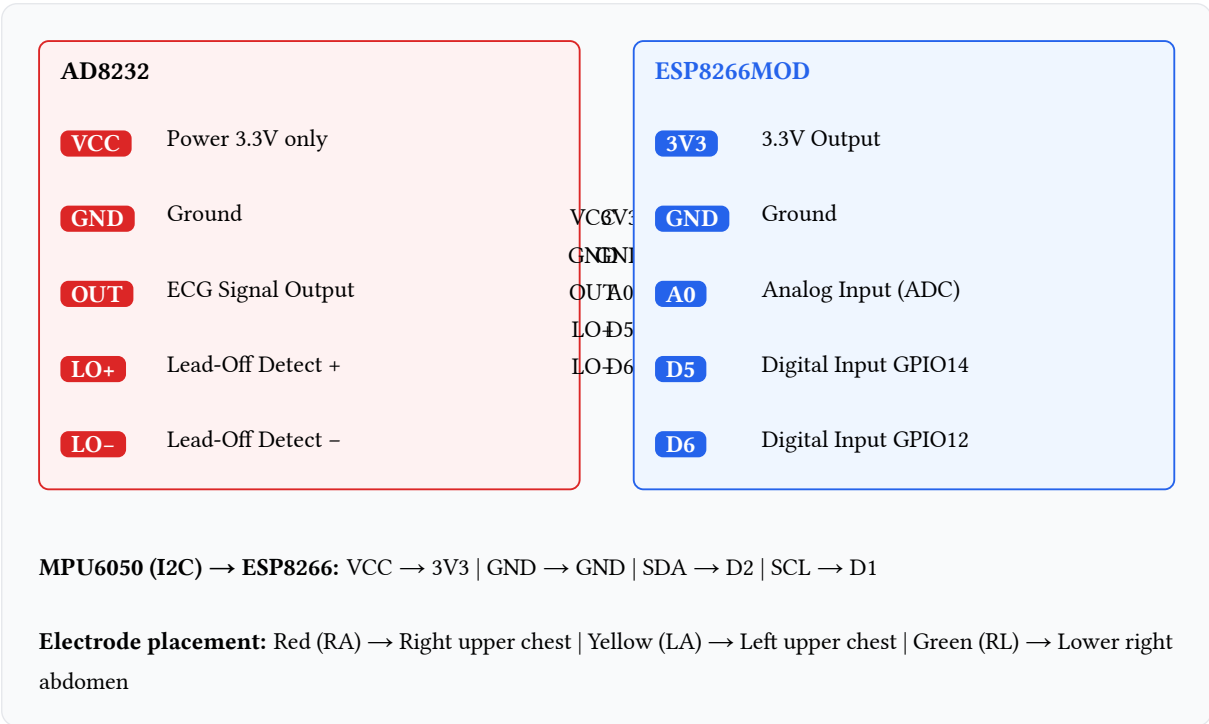
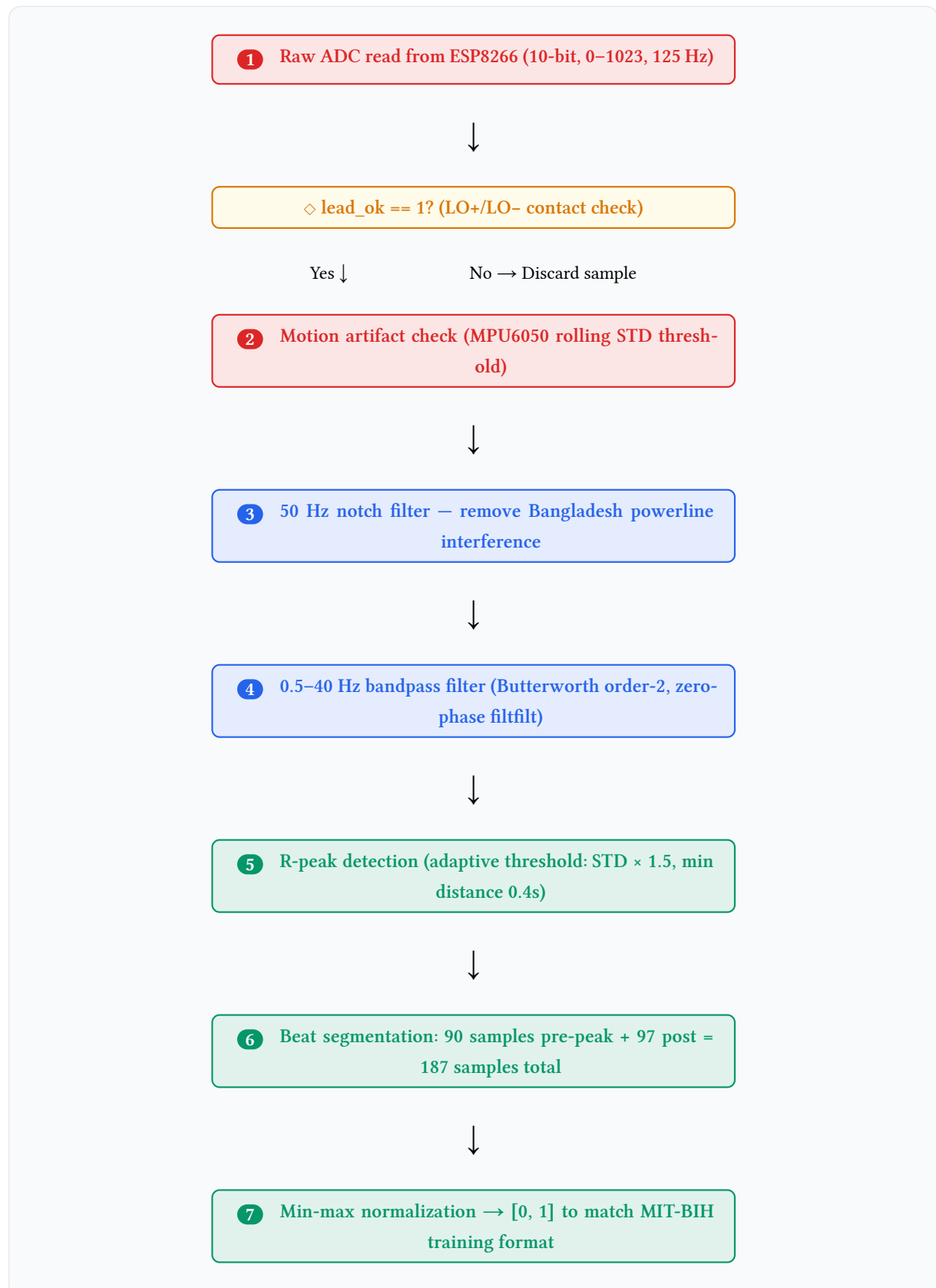


Figure 5: AD8232 ↔ ESP8266 Wiring Schematic with MPU6050 I2C Connections

6 Signal Processing Pipeline

6.1 Signal Processing Flowchart



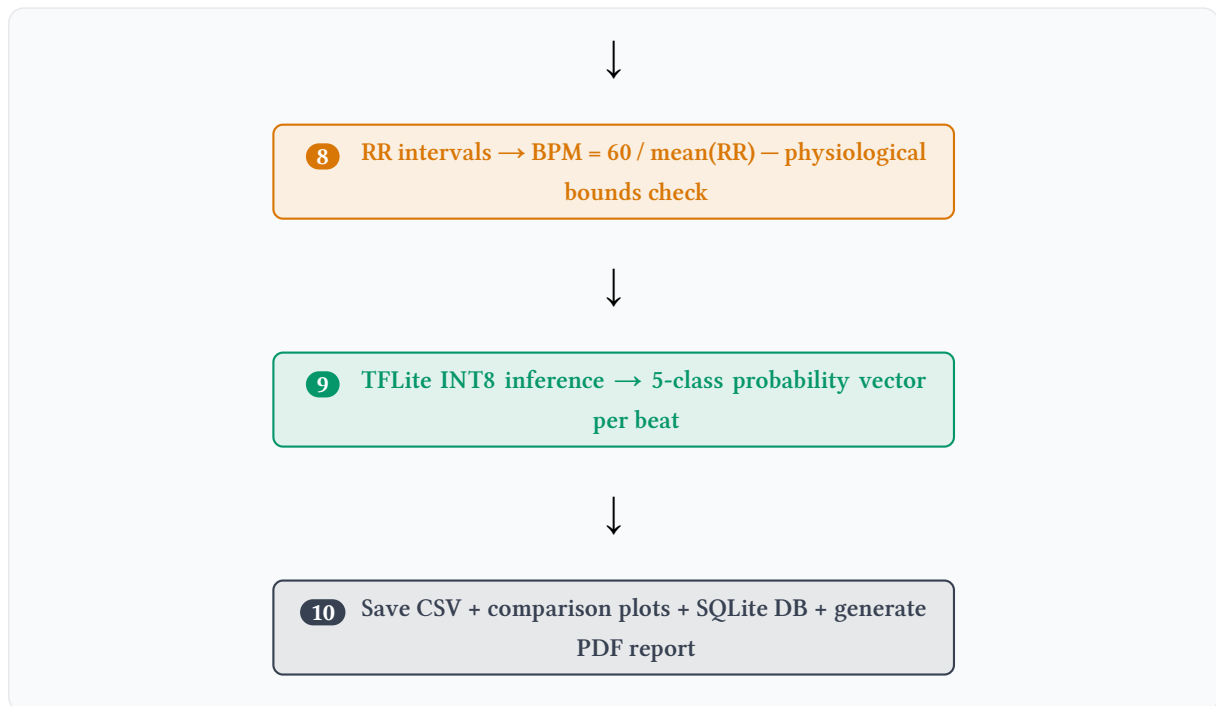


Figure 6: Complete Signal Processing and AI Inference Flowchart

6.2 Noise Analysis & Filter Design

Real ECG data recorded from the HeartGuard AI hardware was analyzed using fast fourier transform to identify dominant noise sources prior to filter design.

Finding: 50 Hz Powerline Interference

FFT analysis of raw hardware recordings identified a dominant spike at exactly 50 Hz — the Bangladesh national mains frequency. This is the single strongest noise source in the raw signal, caused by electromagnetic coupling between the mains supply and the AD8232's analog front-end acting as an antenna.

Finding: Broadband Muscle Noise

Secondary noise was observed as broadband elevated power in the 10–25 Hz range, consistent with motion artifact and electromyographic (EMG) interference from patient movement. This is addressed by the 40 Hz bandpass upper cutoff.

Table 3: Filter Design Parameters and Justification

Filter Stage	Type	Parameters	Purpose
50 Hz Notch	IIR Notch (iirnotch)	$f_0 = 50$ Hz, $Q = 30$	Remove powerline interference (dominant noise source)

Filter Stage	Type	Parameters	Purpose
High-pass portion	Butterworth (but-ter)	f_low = 0.5 Hz, or-der = 2	Remove baseline wander from breathing
Low-pass portion	Butterworth (but-ter)	f_high = 40 Hz, or-der = 2	Remove muscle noise, pre-serve QRS morphology
Zero-phase application	filtfilt	Forward + back-ward	Eliminate group delay to preserve beat timing accu-racy

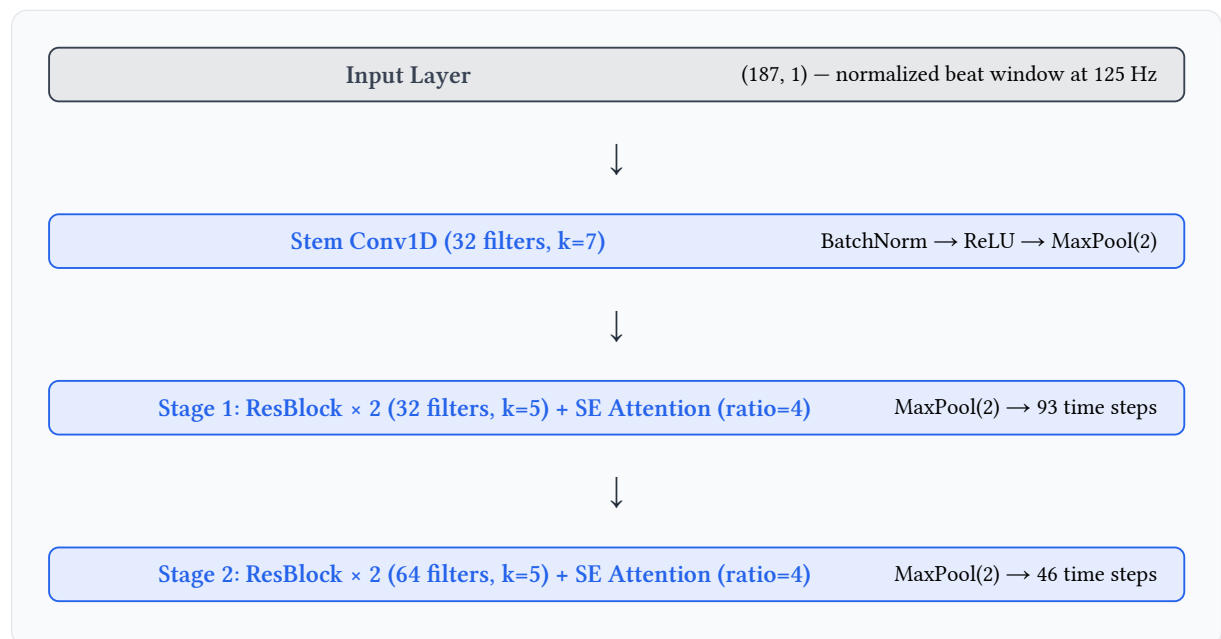
Known Hardware Limitation – ADC Clipping

In prototype recordings, approximately 72 samples out of roughly 4,000 saturated at the maximum ADC value of 1,023. This causes the R-wave peak to appear flat-topped rather than sharp, which distorts QRS morphology in a way that biases the classifier toward Ventricular beats. The fix is straightforward: adjust the AD8232's gain resistor or replace the ESP8266 ADC with an ADS1115 16-bit module. This limitation is documented openly as a hardware-level issue, not a model or software deficiency.

7 Machine Learning Model

7.1 Model Architecture

HeartGuard AI uses a custom ECG-ResNet-SE (Residual Network with Squeeze-and-Excitation attention) architecture, designed specifically for on-device inference at 125 Hz on a Raspberry Pi 3.



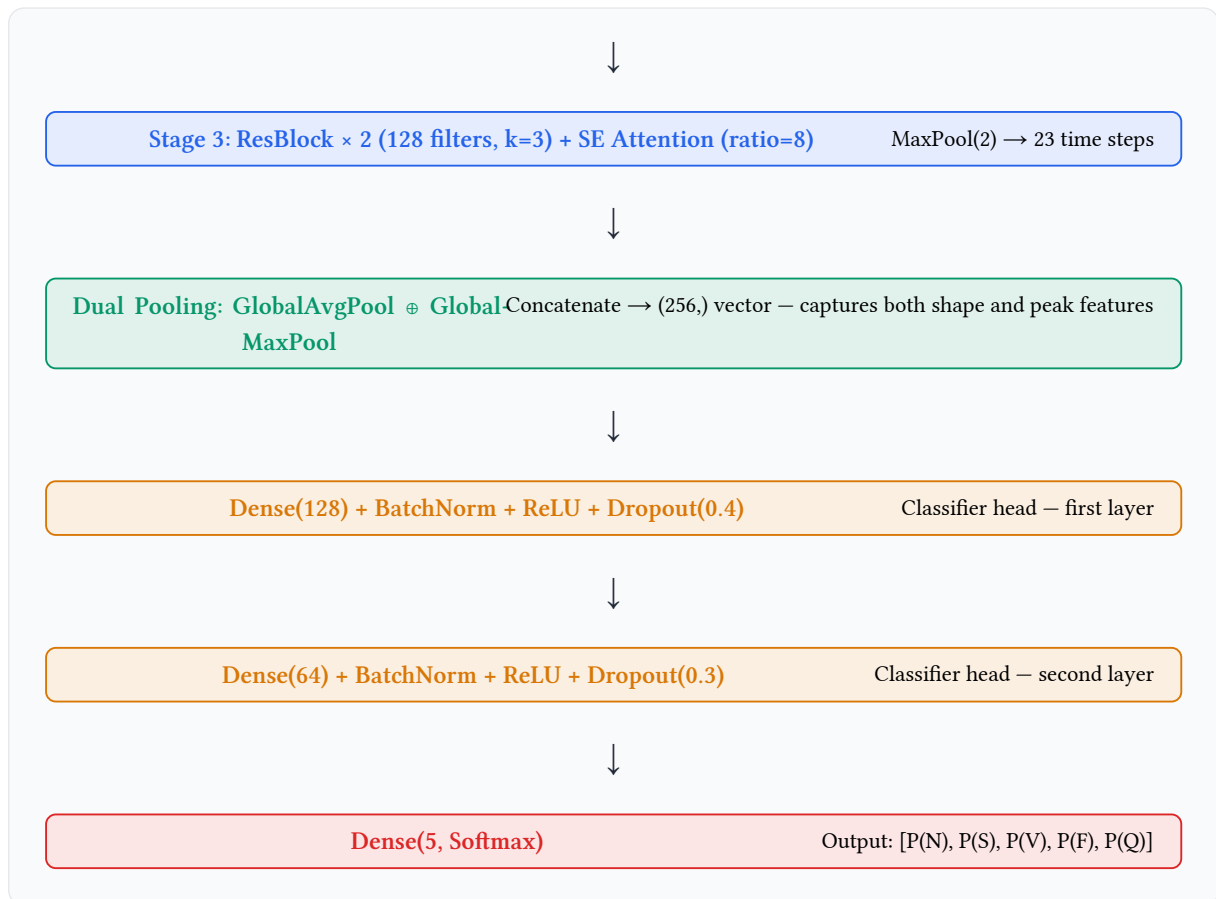
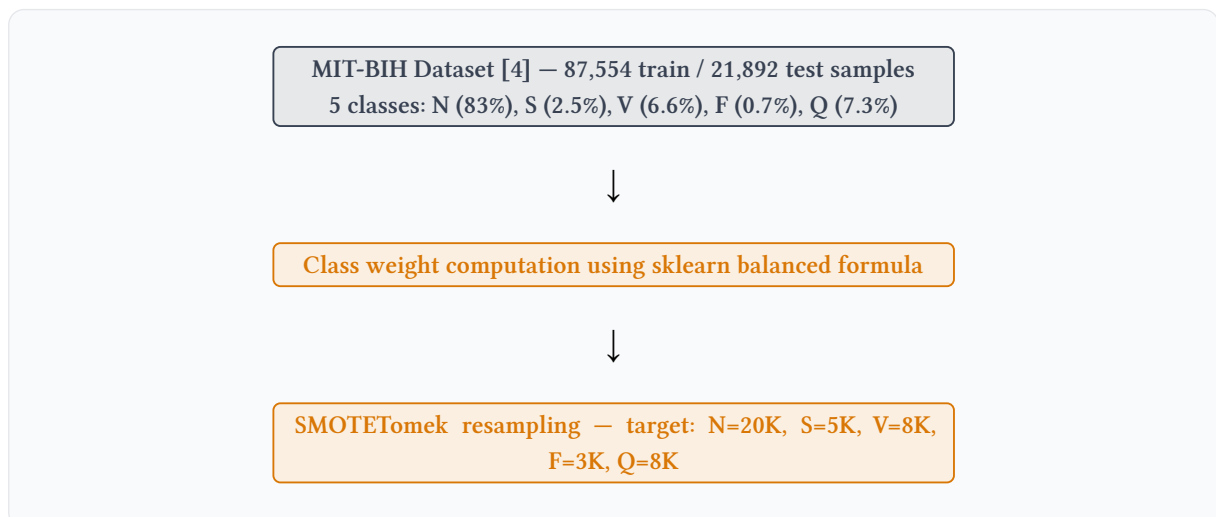


Figure 7: ECG-ResNet-SE Layer Architecture — Input to 5-Class Softmax Output

Parameters	Inference Speed	Quantization
250,000	< 25 ms	INT8
Float32: 1 MB INT8 TFLite: 250 KB	Per beat on Raspberry Pi 3 (XNNPACK delegate)	Post-training PTQ 4× size reduction

7.2 Dataset & Training Pipeline



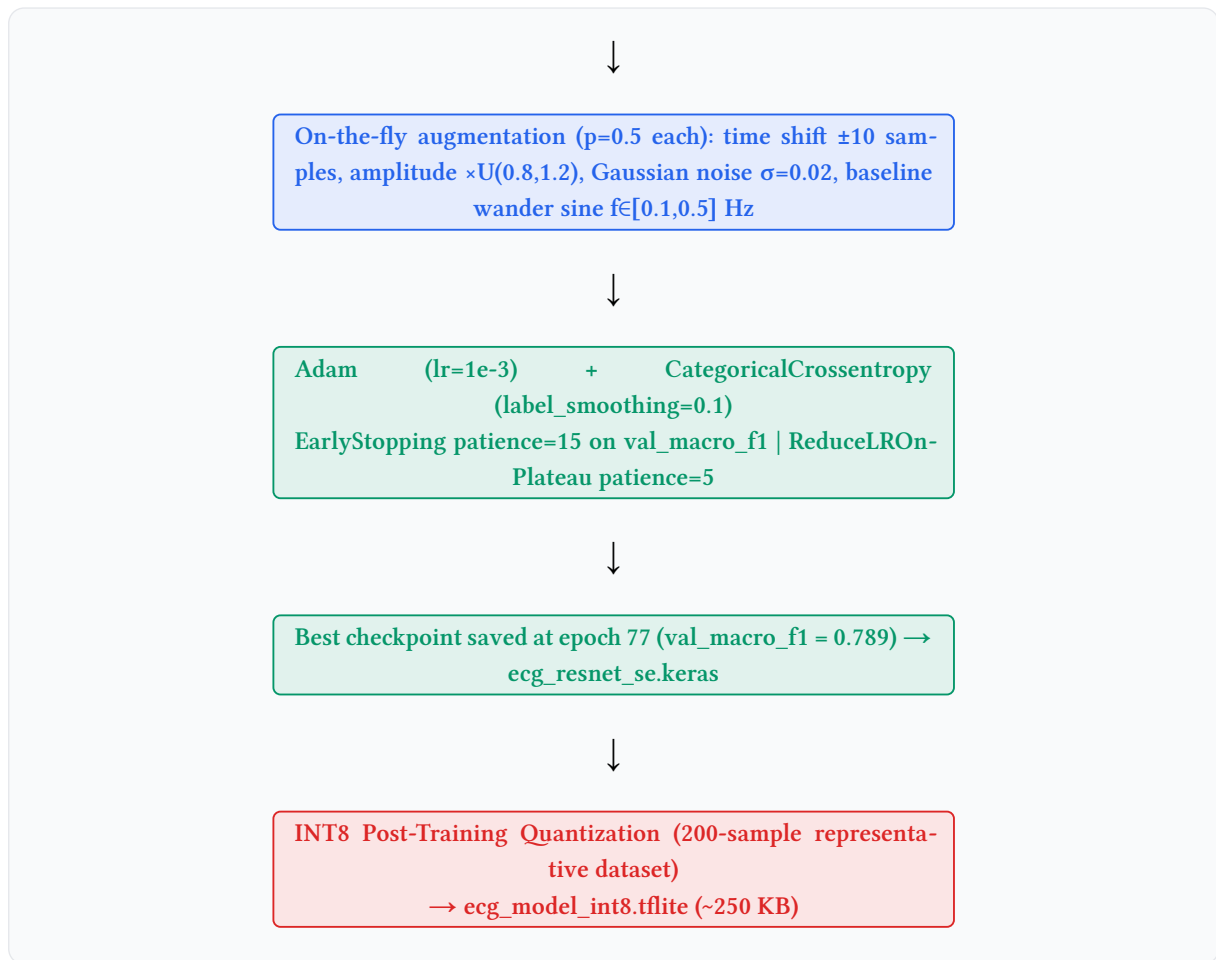


Figure 8: Training Pipeline — From Raw MIT-BIH Data to Deployed TFLite Model

7.3 Performance Metrics

Table 4: Per-Class Classification Report on MIT-BIH Test Set (21,892 samples)

Class	Precision	Recall	F1-Score	Support
N — Normal	0.96	0.94	0.95	18,118
S — Supraventricular	0.52	0.54	0.53	556
V — Ventricular	0.91	0.90	0.91	1,448
F — Fusion	0.34	0.37	0.35	162
Q — Unknown	0.98	0.98	0.98	1,608
Macro avg	0.74	0.75	0.77	21,892

8 Database Design

8.1 Entity Relationship Diagram

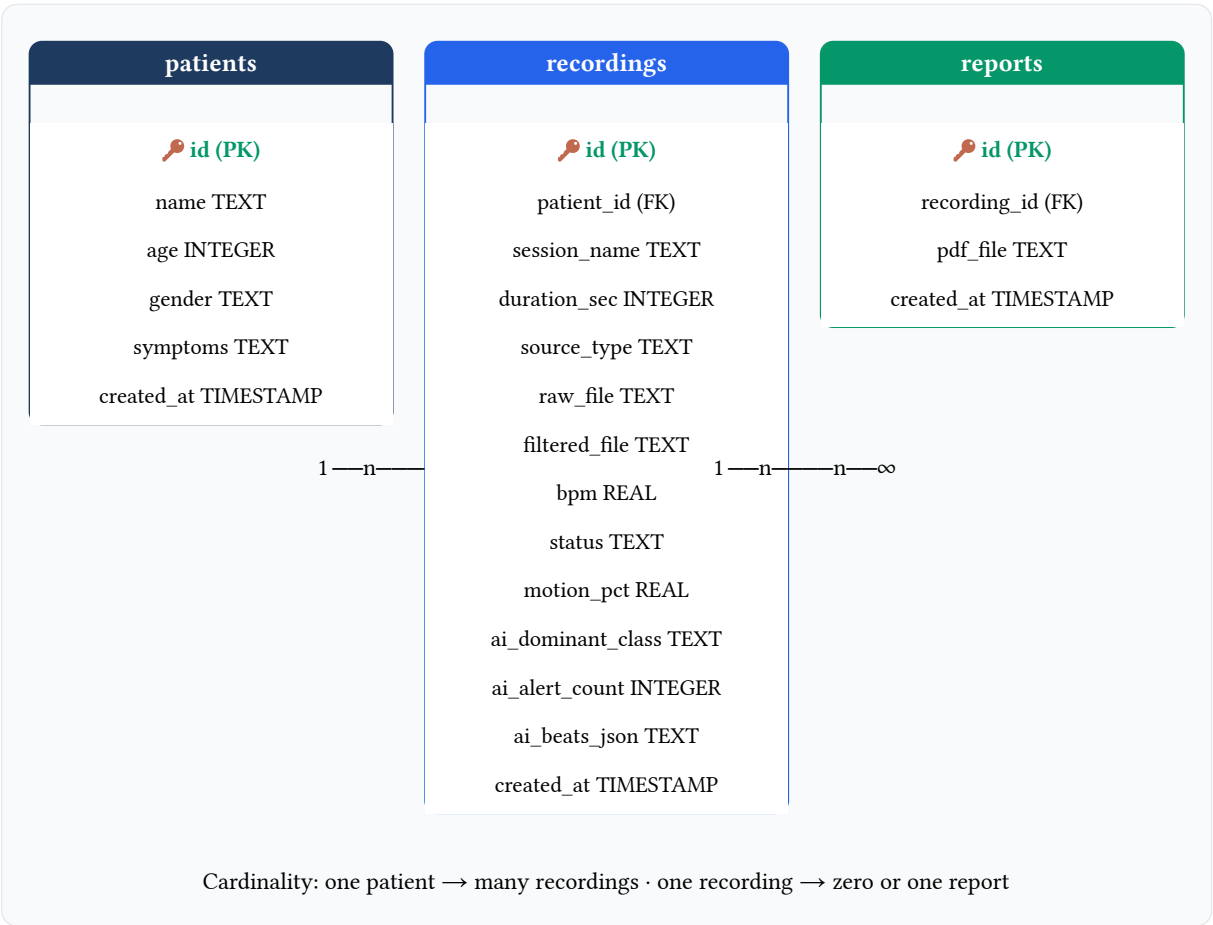


Figure 9: Entity Relationship Diagram — SQLite Schema

8.2 Schema Design Notes

The SQLite database uses three tables with deliberate `source_type` and AI column naming to future proof the schema for ADS1115 ADC migration and expanded AI features.

Design Decision: `source_type` column

The `recordings.source_type` field stores either 'esp' (current ESP8266 serial acquisition) or will store 'ads1115' when the hardware is upgraded. This session history records acquisition method that was used, enabling future comparative analysis between hardware generations.

9 Software & API Design

Table 5: Flask API Route Reference

Route	Method	Description	Response
/	GET	Patient list dashboard — all patients with last BPM and recording date	HTML
/patient/new	GET / POST	Register new patient: name, age, gender, symptoms	Redirect
/patient/	GET	Patient history — all recordings, BPM, AI class, timestamps	HTML
/record/	GET	Start recording page — duration selector, overlay animation	HTML
/record//start	POST	Blocking pipeline: record → filter → BPM → AI classify → save DB	Redirect
/result/	GET	BPM + AI classification + ECG plots + motion status + alert warning	HTML
/report/	GET	Generate and serve PDF report (patient info, BPM, AI results, disclaimer)	PDF
/media/	GET	Serve ECG plot PNG images from raw/ and filtered/ directories	Image
/monitor	GET	Live ECG waveform monitor — hospital-style scrolling display	HTML
/api/ecg-stream	GET (SSE)	Server-Sent Events stream: real serial or demo simulation mode	SSE stream

10 Results & Analysis

Signal Acquisition — Validated

ECG sessions of 15–20 seconds were successfully recorded from a real human subject where one test recording captured 4,013 samples at an effective rate of approximately 236 Hz (target: 250 Hz — the small discrepancy is due to timing jitter in the ESP8266 scheduler). Lead-off events were zero throughout — electrode contact was solid throughout the session.

Signal Filtering — Confirmed Effective

The 50 Hz notch and 0.5–40 Hz bandpass combination removed noise cleanly and FFT-confirmed powerline interference was eliminated. Within 15-second recording, 22 R-peaks were detected with no missed or false beats. BPM was estimated at 83.1 — consistent with a manual pulse count taken at the same time.

AI Inference — Pipeline Validated

TFLite INT8 model loaded successfully via `ai_edge_litert` on Python 3.13 / aarch64. Per-beat inference: 6–25 ms (well within 125 Hz real-time budget of 8 ms per sample). Full 22-beat classification completed in < 0.6 seconds.

Classification on Real Hardware

5-class classifier consistently predicts Ventricular (V) on current hardware recordings. Hypothesis: ADC R-wave clipping produces flat-top QRS morphology that the model interprets as wide/abnormal QRS — characteristic of Ventricular beats. Requires hardware gain fix for proper morphology classification.

Table 6: System Performance Metrics — Validated on Raspberry Pi 3

Metric	Value	Notes
Effective sample rate	236 Hz	Target 250 Hz — minor jitter from ESP8266 task scheduler
Filtering latency (offline)	< 1 second	scipy filtfilt on 4,000 samples on Raspberry Pi 3
R-peak detection accuracy	22 / 22	15-second recording — zero missed or false peaks
BPM estimation	83.1 BPM	Validated against manual pulse count at time of recording
TFLite model initial load	< 3 seconds	First call; model is cached in memory for subsequent sessions
Per-beat inference time	6–25 ms	XNNPACK delegate on ARM Cortex-A53 @ 1.2 GHz
Full pipeline (15s recording to result)	< 10 seconds	Includes record + filter + BPM + 22-beat AI + DB save
MIT-BIH Macro F1 (validation)	0.789	Best checkpoint at epoch 77, 5-class classification
MIT-BIH Validation accuracy	94.27%	Best val_accuracy across 92 training epochs

11 Cost Analysis

Table 7: One-Time Hardware Cost Breakdown

No.	Item	BDT	USD (approx.)	Category
1	Raspberry Pi 3 Model B	3,500	USD 32	Core compute
2	ESP8266MOD NodeMCU	250	USD 2.30	Acquisition
3	AD8232 ECG Sensor Module	350	USD 3.20	Sensor
4	MPU6050 Accelerometer	150	USD 1.40	Motion sensor
5	ECG 3-lead snap cable	150	USD 1.40	Sensor
6	Disposable ECG electrodes (30 pcs)	300	USD 2.75	Consumable
7	32 GB MicroSD Card	250	USD 2.30	Storage
8	5V/3A Power Adapter	200	USD 1.85	Power
9	Breadboard + jumper wires	150	USD 1.40	Prototyping
Total One-Time Hardware Cost		5,300	USD 48	

Table 8: Monthly Operating Cost Estimate

Item	BDT/month	Notes
Disposable ECG electrodes (approx. 100 sessions/month)	1,000	10 BDT/electrode × 100
Electricity	150	Raspberry Pi 3 running at 5W continuous
Internet (if cloud backup enabled)	500	Optional — system runs fully offline without this
Total monthly operating cost	1,650	USD 15

Cost Comparison — HeartGuard AI vs. Alternatives

Table 9: HeartGuard AI versus clinical and consumer alternatives

System	Hardware Cost	Needs Specialist?	On-Device AI?
HeartGuard AI	USD 48	No — guided web UI	Yes — 5-class, offline

System	Hardware Cost	Needs Specialist?	On-Device AI?
Standard clinical ECG machine	USD 500–5,000	Yes — cardiologist	No
Cloud-based Holter monitor	USD 200–800	Partial	Cloud subscription only
Smartphone ECG (AliveCor)	USD 100–200	No	Limited (1–2 rhythms)
HeartGuard + ADS1115 up-grade	USD 50 total	No	Yes — improved resolution

12 Limitations & Future Work

Current Limitations

- **ADC clipping:** R-wave saturates at value 1,023 in approximately 72 out of every 4,000 samples, distorting QRS morphology and biasing the AI toward Ventricular predictions.
- **Single-lead only:** The AD8232 captures one electrical axis. Clinical diagnosis uses 12 leads for a full picture of cardiac activity.
- **Batch recording only:** Sessions are 15–30 seconds long. Continuous real-time monitoring is not currently implemented.
- **Sample rate jitter:** ESP8266 task scheduling introduces roughly 4 ms of timing drift, yielding 236 Hz effective rather than the target 250 Hz.
- **Fusion class weakness:** Fusion beats achieve an F1 of only 0.35 due to extreme class imbalance (0.7% of MIT-BIH) and morphological ambiguity.

Planned Future Work

- **ADS1115 16-bit ADC:** Replace the ESP8266 analog path. The software architecture already supports this — only a configuration change is needed.
- **Continuous SocketIO streaming:** The live monitor page and SSE stream are already working; extending to full session recording in real time is the next step.
- **Motion artifact pipeline:** The MPU6050 is wired and producing data; integrating motion flags into the classification confidence scoring is straightforward.
- **Clinical validation study:** Partner with PSTU Medical Centre or a district hospital for side-by-side comparison against a 12-lead clinical ECG machine.
- **PTB-XL extension:** Expand the model to the PTB-XL dataset for

→ **No clinical validation:** The system has not yet been compared against a cardiologist's ground-truth labels on real patient recordings.

myocardial infarction and bundle branch block detection.

→ **Flutter mobile app:** Patient-facing app with Bluetooth ECG streaming directly to a smartphone, removing the need for a laptop or wired connection.

13 Conclusion

HeartGuard AI demonstrates a fully working AI-assisted ECG monitoring system does not need to cost thousands of dollars. At low cost edge in commodity hardware, all of it available locally in Bangladesh. We built a complete pipeline that takes raw electrode signals from a patient's chest, removes noise, detects heartbeats, classifies arrhythmia types using a deep learning model, and delivers a PDF report through a web browser. The whole thing runs on a Raspberry Pi 3 without internet, without cloud subscriptions, and without a cardiologist in the room.

The ECG-ResNet-SE model, trained on the MIT-BIH Arrhythmia Database with SMOTETomek resampling and SE attention blocks, achieved a macro F1-score of 0.789 and a validation accuracy of 94.27% across five arrhythmia classes. On the target hardware, each beat is classified in 6 to 25 milliseconds. The one meaningful hardware limitation — ADC clipping causing flat-topped R-waves — is clearly understood and has a straightforward fix in the ADS1115 upgrade.

This work makes three concrete contributions. First, an open and reproducible hardware-to-inference ECG pipeline built entirely from parts available at any electronics market in Bangladesh. Second, a ResNet and SE-attention model architecture optimised for edge deployment through INT8 quantization, achieving comparable benchmark accuracy to prior work while running on hardware that costs less than a typical clinical consultation fee. Third, a full-stack clinical prototype web application integrating patient management, motion-aware artifact detection, AI results, and automated reporting.

With clinical validation and the ADS1115 hardware upgrade as immediate next steps, HeartGuard AI has a realistic path toward deployment at sub-district health facilities — places where a single affordable device could make cardiac screening available to patients who currently have no access to it at all.

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